

Conversion of data into visualizations to draw valuable insights Transcript

Welcome back, now we will be using the encoding features to design effective visualisations. By effective I mean your audience should be able to discharge three functions that is they should be able to discriminate, rank and estimate the data that is there in the visual and now we will see what are these seven features that are available to you to create visuals that can effectively discharge these functions of discrimination, ranking and estimation. These are in order in the decreasing order of preference with position on common aligned scale being the highest with colour being the lowest.

These features are generally available in all visualisation softwares. It is up to the visualisation creator or the analyst to choose them scientifically and strategically so that the information on the visual can be easily discriminated, ranked and estimated by the audience. We will now take examples for each one of these we are encoding features and see how they will improve the visualisation.

Now a caveat here or we would like to add I would like to add that the examples are given directly in the lecture this will be followed by a hands-on experiential learning session where you will be trying to test these encoding features and improve the effectiveness of your visualisation. Now let us see the examples for each of these encoding features. First we will start with the colour which is lowest in the reference order because it has both advantages and many disadvantages as respect to the discriminate rank and estimate function.

On the right side you see the 2019 Lok Sabha elections voters margins by constituencies in Andhra Pradesh. This graph is presented as a hundred percent stacked chart. We are not getting into the choice of the chart we are just trying to showcase the ability of an encoding features to help you do the discriminate rank and estimate functions.

Now in this graph we are simply plotting in a decreasing order of the vote share from highest to lowest the constituency and we are just presenting the vote margin and it is has to be read with the help of the colour saturation palette and it is the legend is given here. So if you see it is used dark brown for constituency with highest votes where the margin is close to 0.57 million that is roughly 5.7 lakhs. Now you have at the lowest end roughly about 30,000 votes or 0.37 3,000 votes where you have the lowest end in light blue a light brown colour.

The graph per se does not clearly indicate the information that you want to convey. What is the information that we want to convey? We want to convey strikingly when we see the graph the audience should be able to tell which constituency had the highest voter margin or winning margin for the candidate. So now this information is not visible in the graph.

However I have used colour to show that some amount of distinction between the

constituencies which had the highest voter share or highest voter margin for the winning candidate as compared to other constituency. So you are able to in some sense discriminate but can you rank them clearly? You can yes to an extent you can rank them clearly but not very that is not coming up front then that is not coming very clearly when you see the graph. You have to put an effort to understand the ranking.

Discrimination to an extent you are able to do immediately whereas estimation part is absolutely missing from this. You cannot say by what factor Kakinada or Kadapa are different. Okay the Kadapa constituency had highest margin winning margin as compared to Kakinada which has least margin as per the graph.

Just that you can say highest and least but you cannot say by what factor or by how much. Therefore the graphic though using colour as an encoding feature fails to perform the ranking and the estimate functions to an extent is it is able to do the discriminate function. Now let us see to what is the next level of encoding feature that is available to you to improve your visuals.

This is volume. In this case you will be using a bubble chart or a modified bubble chart in which the area of the circle indicates the voter margin or votes margin or the winning margin of the candidate in that particular constituency. You are able to clearly discriminate by looking at the size of the circle and you are also able to do a ranking saying that this is particular constituency has higher winning margin as compared to the other constituency.

However estimation we have to put little more effort in terms of comparing the size of the circles. It is easy to say that Hisar is almost 5 times of Faridabad is almost 5 times of Hisar given the size of the circles but again there is some amount of effort that you have to put in comparing circles. If somebody asks Gurgaon and Karnal you can say they are almost similar but you know Gurgaon is little lower than Karnal but you cannot tell by the factor.

All that you can say if it is little lesser or little higher you have to use these adjectives but not give a clear estimate. This is serving a function better than the colour but it is not helping you to give all the three objectives fully that is discriminate. Yes it is able to clearly discriminate.

It is also help to rank but that also not fully. Again for estimate it is helping but not fully. So this graphic using of volume is better than using of colour because it is able to communicate the same information in a much faster or efficient manner.

Again I am reminding you the objective here is to enable our audience to perform the discriminate rank and the estimate functions using this visual cues on your graphic. Then we come to the other encoding feature called angle. Here we take the help of line graphs.

We try to simply show what is the vote margin of the select constituencies or winning margin of the select constituencies. Here again for simplicity I have taken three constituencies Tirupati, Vijayawada and Visakhapatnam 2019 blocks of elections. So we can see the vote share of Visakhapatnam.

Tirupati has fallen between 2004 to 2009 then again it has shot up in 2019 whereas Visakhapatnam and Vijayawada have seen sharp fall in the winning margins. Now here you are able to discriminate by looking at 2019 line saying that Tirupati has very high winning margin. By looking at the graph you can simply say that Tirupati has very high winning margin as compared to Vijayawada and Visakhapatnam.

So you are able to immediately discriminate and you are also able to rank them Tirupati is greater than Vijayawada and Vijayawada is almost equal to Visakhapatnam given that they are in the same region. Now as far as estimation comes this is where it becomes a little tricky to immediately draw certain conclusions where was the highest growth in the winning margin. This kind of question is not answered by this graph.

So for that you have to plot the rate of change in the vote share or the winning margin in these constituencies which is given in the graph on the extreme right. So here you can clearly see Tirupati has seen a secular increase in the winning margin in the last two elections whereas Visakhapatnam has seen a moderation Vijayawada has seen a swing in terms of increase in winning share then it has fallen. So this type of conclusions can be drawn depending upon what kind of metric are you using.

So again it points out to the question what is the information that you want to convey to your audience. In this case using the rate of growth is much more beneficial than using a simple line graph. You can also show the rate of increase of the vote margin or winning margin.

We now come to the last two important features called length and position. So the length of the geometric object reflects the quantity that you want to reflect. So finally once you come to the most effective way of presenting this you can do a horizontal or vertical bar which we have been doing in our earlier classes and rank the constituencies in terms of the vote margin from decreasing to increasing from increasing to decreasing margin and clearly show a comparison between Kadapa and Anakapalli or Machilipatnam for that matter.

We can simply do the discrimination function we can clearly bring vote which constituencies are equal or not equal. The person can immediately look at for example here in Anantapur and Hindupur one can hear clearly say that these two are almost similar or there is no much difference in the vote share because the length of the bar is clearly representing the vote margin which is almost similar though Anantapur is little higher than Hindupur. Then you also can do the ranking function saying that Kadapa is highest compared to Guddu which is the lowest.



Then you can also estimate the times by which you can the times by which the vote margins vary. And finally the most effective way is to do it is when it is positioned against the common axis. So in this case you are having the points which indicate the vote share.

So you can measure the distance and clearly see Faridabad has the highest vote share as compared to Ambala and the distance can be easily compared for making the estimation function right. So by using the different principles like using colour saturation, volume, angle, slope we can create more levels of effectiveness into our visuals and enable our audience to do the discrimination ranking and estimate function. Finally the main purpose of our visualisation is it has to be expressive and effective.

Expressiveness represents how well the visualisation is showing all the facts in the data set right. How expressive is your visual that it covers the major portion of the data that you want to express. Then effectiveness is you should be able to do it in the most optimal fashion without there is scope for further improvement.

Information can be said effectively presented if you cannot improve it further. So it is most readily perceived compared to any other form of presentation. So on the graph on the slide you see a comparison of different types of variables that you have the quantitative variable, ordinal data, nominal data and different types of encoding that you can use to improvise the effectiveness and expressiveness of your visuals.

We will look at this in the live example.

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